Prashanth Duggirala

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PROFESSIONAL SUMMARY

AI Engineer with 5+ years of experience building production AI systems across media intelligence and recruiting automation domains. Specialized in designing multi-agent architectures with LLM-powered orchestration, tool development, and evaluation frameworks. Deep expertise also spans core ML; data curation, model training, and deploying scalable inference pipelines in cross-functional environments.

EXPERIENCE

Meltwater, Hyderabad — *AI Engineer III*

DEC 2025 - PRESENT

- **Architected and implemented a multi-agent routing system** for Meltwater's AI-powered media intelligence chatbot, designing an LLM-driven agent router on LangGraph that dynamically dispatches queries across specialized sub-agents (content tracking, brand analytics, news summarization) — enabling multi-turn conversational workflows with phase-gated execution, async tool calling, and stateful conversation management, achieving **97% routing accuracy**.
- **Developed domain-specific AI agent tools** (media monitoring, journalist discovery, press release impact analysis, coverage reporting) exposed via Model Context Protocol (MCP) on a LangChain ReAct architecture — reducing average **query resolution time by 50%** across Meltwater's media intelligence platform.
- **Extended a custom LLM evaluation framework** (DeepEval, DSPy, LangSmith) with unit and end-to-end evals covering citation accuracy, routing correctness, and adversarial test cases — achieving **>90% eval coverage**. Delivered platform-wide backend improvements to the FastAPI chatapi including intent classification and robust error handling with Prometheus and Grafana monitoring, **cutting targeted error-type rates by 80%**.

Leoforce Inc., Hyderabad — *Senior Data Scientist*

JUNE 2021 - DEC 2025

- **Architected IRA, a chat-based AI recruiting agent** powered by LLMs, hybrid search (dense + sparse retrieval), and intelligent agent workflows with RAG-based dynamic few-shot selection — automating job matching, resume screening, and real-time candidate engagement at scale, achieving a 75% automation rate and **reducing manual screening time by 50%**.
- **Engineered an end-to-end NLP pipeline** combining custom parsers, domain-adapted BERT for NER, and LLM-based data augmentation to extract structured insights from **500K+ daily** unstructured documents — achieving **90% parsing accuracy and 92% F1-score**. Built scalable inference pipelines on Docker/Kubernetes (EKS) with MLflow tracking and real-time Grafana dashboards for proactive model monitoring.

PROJECTS

Text-Prompted Image Segmentation for Drywall Quality Assurance

- Developed and fine-tuned a text-conditioned segmentation model (**CLIPSeg**) using a discriminative learning rate strategy to identify drywall cracks and taped joints from natural language prompts.
- Achieved an overall **mIoU of 0.55** and Pixel Accuracy of **0.94**, enabling **zero-shot** and referring expression segmentation for both defect classes handled by a single model, with a robust post-processing pipeline for production-ready binary masks.

Generative Adversarial Framework for Eye Image Synthesis and Gaze Estimation

- Developed DCGANs for synthetic eye image generation, producing high-fidelity images across multiple gaze directions and lighting conditions that improved **gaze estimation accuracy by 35%**.
- Created a **custom evaluation framework** using pre-trained classifier loss to quantify image realism, reducing data collection requirements and establishing a benchmarking methodology.

SKILLS

Machine Learning: LangChain, Agents SDK, DSPy, LLM, VLM, ViT, MLFlow, Python, PyTorch, HuggingFace, Pandas, Numpy, Copilot, OpenAI, Bedrock

Databases: Elastic, MongoDB, MySQL

Web: React, JavaScript, CSS, OAuth, Streamlit, Appsmith

DevOps: Git, Docker, Kubernetes, AWS, GitLab CICD, EC2, OpenSearch, EKS

EDUCATION

University of California, Davis— *M.S. Computer Science, 3.96/4*
SEPTEMBER 2021

Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram — *B.Tech. Computer Engineering, 8.92/10*
JULY 2019

PUBLICATIONS

Liaqat, S., Wu, C., **Duggirala, P.R.**, Cheung, S.S., Chuah, C.N., Ozonoff, S., Young, G — ***Predicting ASD diagnosis in children with synthetic and image-based eye gaze data***, Signal Processing: Image Communication, Vol. 94, 2021. DOI